

Applied Math in Society

MATH 260

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Contents

Course Documents	1
Course Notes	2
LLMs and Dimensionality.	2
In-Class Activities	8
Human Data Workers	9
PCA and <i>t</i> -SNE by Hand: Two Toy Examples	10

Course Documents

Course Notes

The following notes are intended to remind you what we covered each week. They are not a substitute for attending class and not a substitute for reading the textbook.

LLMs and Dimensionality

This is an outline of (some of) the topics we covered in class.

Large Language Models and Dimensionality AI Ethics and the Global Data Supply Chain: A Consolidated Guide

The training corpora for modern language models are huge. For example, the training corpus for GPT-3 contains 300 billion tokens. The rule of thumb for tokens-to-words conversion is that 1 token corresponds to about 0.75 words, so the training corpus for GPT-3 contains about 225 billion words. If this were a book, Wolfram Alpha estimates that it would be about 308 million pages, which would talk about 1600 years of silent reading (2900 years in audio book format, and 7100 years if you were to try to type it out).

BERT, from 2018, was trained on a corpus of 3.3 billion words from the English Wikipedia, which is about 0.0015% of the size of the GPT-3 training corpus. Larger models rely on illegally scraped data, which is not publicly available, so we don't know the exact size of their training corpora. But we do know that large libraries, like Library of Congress, contain about 17 million books, which is about 10 billion words, so the training corpora for large language models are at least an order of magnitude larger than the entire Library of Congress.

The training of large language models is also very expensive. For example, the training of GPT-3 is estimated to have cost about 4.6 million dollars, and the training of GPT-4 is estimated to have cost about 100 million dollars. The training of large language models also requires a lot of computational resources, which are not accessible to most researchers and practitioners. For example, the training of GPT-3 is estimated to have required about 355 years of GPU time, and the training of GPT-4 is estimated to have required about 10,000 years of GPU time.

The human cost of training large language models is also very high, as it requires a lot of human labor to collect and preprocess the training data, and to fine-tune the models for specific tasks. For example, the training of GPT-3 is estimated to have required about 10,000 hours of human labor, and the training of GPT-4 is estimated to have required about 100,000 hours of

human labor. This labor includes tasks such as data collection, data cleaning, data annotation, and model evaluation, which are all essential for the successful training of large language models. Filtering for graphic or inappropriate content is also a significant part of the human labor involved in training large language models, as it is important to ensure that the training data does not contain harmful or offensive content that could be learned by the model and reproduced in its outputs.

This section explores the ethical implications of artificial intelligence, ranging from college campus algorithms to the global labor force. Discussion questions and insights based on the Pulitzer Center investigation into the “Platform Proletariat” in Brazil.

The Army of Workers.	The “magic” of AI is an illusion. It does not appear “magically” through machines; it requires an army of workers to comb through prodigious amounts of data to teach systems what is acceptable.
Modern Assembly Line.	Like the Industrial Revolution, workers often perform repetitive “microtasks” (tightening a bolt) without knowing if they are training systems for surveillance and repression.
Labor Rights and Inequality in the Global South.	The AI data supply chain relies on underpaid workers in countries like Brazil rather than Silicon Valley offices.
Precarious Work.	Workers are paid pennies for complex tasks and face issues like non-payment and strict non-disclosure agreements that prevent them from speaking out about their working conditions.
Addictive Platforms.	Some platforms embed microtasks that addict children and teens who work for hours for very little pay.
Quality vs. Accuracy.	Poor working conditions directly impact the quality of AI outputs.
The Opaque Chain.	The chain of command is intentionally secretive, involving tech giants, intermediary platforms (like Appen), and isolated workers

Principal Component Analysis (PCA) and t -Stochastic Neighbor Embedding (t -SNE)

Principal Component Analysis (PCA) and t -distributed Stochastic Neighbor Embedding (t -SNE) are both methods for reducing the dimension of a dataset. However, they preserve different kinds of structure.

In this worksheet, we compute PCA exactly for a small dataset and then approximate the behavior of t -SNE using neighbor probabilities.

PCA identifies directions of maximum variation in a dataset. It finds the directions in which the data spreads out the most and projects the data onto those directions to reduce its dimension. The linear algebra required to make this happen? Eigenstuff! This allows us to take linear combinations of the “features” (the measured properties) to create new variables, where fewer variables are needed to explain most of the behavior.

Definition 1 When working with data, we are often interested in measures of central tendency and measures of spread. Informally, these measures help us answer the questions, “Where is the middle of the data?” and “How spread out is the data?” Let X be data matrix where each column corresponds to a feature (or variable) and each row corresponds to an observation (or value). We refer to a column of X as a feature vector, denoted $\bar{x}$. Some measures of central tendency include

1. The **mean** for a feature x : $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.
2. The **median** for a feature x : the value that splits the data into two equal halves.
3. The **mode** for a feature x : the most frequently occurring value.
4. The **midrange** for a feature x : the average of the minimum and maximum values.

Some measures of spread include

1. The **variance** for a feature x : $\text{Var}(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$.
2. The **standard deviation** for a feature x : $\text{SD}(x) = \sqrt{\text{Var}(x)}$.
3. The **covariance** of two features x and y : $\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$.

These measures of central tendency and spread are important for understanding the structure of a dataset and for performing dimensionality reduction techniques like PCA. Sometimes we do some preprocessing to make the data easier to work with. Two common transformations are centering and standardizing a feature. These transformations can be expressed in terms of the mean and standard deviation of a feature.

1. **Centering a feature**: replacing x_i with $x_i - \bar{x}$ for each $x_i \in x$.
2. **Standardizing a feature**: replacing x_i with $\frac{x_i - \bar{x}}{\text{SD}(x)}$ for each $x_i \in x$.

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Checkpoint 2 Prove that, for a centered data matrix X , $\text{Cov}(x, y) = \frac{1}{n} x^T y$.

Checkpoint 3 Consider a data matrix

$$X = \begin{bmatrix} -2 & -1 \\ -1 & -2 \\ 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

where the columns represent two features and the rows represent four observations.

We can interpret the rows of X as points in the plane. The first column gives the x -coordinates and the second column gives the y -coordinates.

1. Plot the four points in the plane.
2. Determine if the data is centered. If not, center it.
3. Determine if the data is standardized. If not, standardize it.
4. Form the standardized and centered data matrix. Verify that the columns of this matrix have mean zero and standard deviation one.

5. Give a geometric description of the data after plotting it in the plane. Does it appear to have a preferred direction of variation?

Checkpoint 4 Before doing any calculations, describe the meaning of each entry of the matrix

$$\frac{1}{4}X^T X.$$

This is called the **covariance matrix** of the dataset.

Confirm by hand that the covariance matrix is

$$\begin{bmatrix} 2.5 & 2 \\ 2 & 2.5 \end{bmatrix}.$$

What does this tell you about the relationship between the two variables?

Checkpoint 5 Find the eigenvalues and eigenvectors of the covariance matrix.

The eigenspaces of the covariance matrix correspond to directions in which the data varies. The eigenvalues give the amount of variation in the direction(s) of the corresponding eigenvector(s). Recall that an eigenspace $E_\lambda = \{v \mid Av = \lambda v\}$. For working with data, we often choose a basis of linearly independent normalized eigenvectors associated to the eigenvalue λ . (Normalized means that the eigenvectors satisfy $\sum_{i=1}^n v_i^2 = 1$.) Even though zero is never an eigenvector, we include it in the eigenspace to ensure the eigenspace is a vector space.

In this case, each eigenspace is one-dimensional, meaning that each eigenspace has a single eigenvector (up to scaling) that forms a basis for the eigenspace.

Checkpoint 6 The largest eigenvalue λ_1 corresponds to the direction of greatest variation in the data. This is called the **first principal component**.

It accounts for

$$\frac{\lambda_1}{\lambda_1 + \lambda_2}$$

of the total variation.

The first principal component term is 4.5 and the first principal component direction (with unit length) is

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Project each data point onto this direction by computing the dot product of each data point with the first principal component direction. This gives a new dataset consisting of one number for each data point, which is the coordinate of the data point in the direction of the first principal component.

1. Compute

$$\frac{x + y}{\sqrt{2}}$$

for each data point in X and draw the projected points on a number line.

2. What structure do you observe in the projected dataset?
3. Explain how PCA has reduced the dataset from two dimensions to one. *t*-SNE.

Using *t*-SNE embeddings, we can visualize high-dimensional data in two or three dimensions while preserving local structure. The key idea is to convert distances between points in the original high-dimensional space into probabilities that represent similarities between points. Then, *t*-SNE finds a low-

dimensional embedding that preserves these similarities as closely as possible.

Informally, a t -distribution is a probability distribution that has heavier tails than a normal distribution. This means that it assigns more probability to points that are far away from the mean compared to a normal distribution, and that's useful when we don't know the true distribution of the data and want to be more robust to outliers.

If you have taken statistics, you may have encountered Student's t test, which is a statistical test used to determine if there is a significant difference between the means of two groups. The t -distribution is named for the work of William Sealy Gosset, who published under the pseudonym "Student" to avoid conflicts with his employer, Guinness Brewery (he held the title of Head Brewer; look what you can do with a math-stats background!) For more fun statistical history, check out the book *How Data Happened* by Chris Wiggins and Matthew L. Jones. Why was Guinness employing statisticians? They wanted to improve the quality of their beer, and the challenge faced by statisticians employed by the brewery was that they had very small sample sizes to work with, which made it difficult to draw reliable conclusions about the quality of the beer. Unlike governments at the time, which used census data (and computing, such as it was!) to gather information about the population, Guinness wanted to save money by using small samples to make decisions about their beer production.

Checkpoint 7 Now consider five points on a number line:

$$A = 0, \quad B = 1, \quad C = 2, \quad D = 8, \quad E = 9.$$

We will approximate how t-SNE measures similarity between nearby points.

Checkpoint 8 Compute the squared distances from point A to each of the other points.

Record:

$$d(A, B)^2, \quad d(A, C)^2, \quad d(A, D)^2, \quad d(A, E)^2.$$

Checkpoint 9 In t-SNE, similarities are converted into probabilities using the formula

$$P(j | A) = \frac{e^{-d(A, j)^2}}{\sum_{k \neq A} e^{-d(A, k)^2}}.$$

1. Compute approximate values for $P(B | A)$ and $P(C | A)$.
2. Explain why $P(D | A)$ and $P(E | A)$ are approximately zero.

Checkpoint 10 Based on these probabilities, which points should appear close to A in a t -SNE plot?

Which points should appear far away?

Sketch a possible two-dimensional arrangement of the five points that preserves these neighbor relationships.

Checkpoint 11 Compare what PCA preserved in the first example with what t-SNE preserved in the second example.

1. What geometric structure did PCA identify?
2. What similarity structure did t -SNE identify?
3. When might PCA be preferable to t -SNE?
4. When might t -SNE be preferable to PCA?

Unlike PCA, t -SNE does not try to find directions of maximum variation. Instead, it tries to keep nearby points close together after reducing the dimension of a dataset.

Friday 8/26

In-Class Activities

These are copies of the in-class activities distributed during the semester.

Human Data Workers

1. **Fact Check Challenge.** Students have exactly 101 seconds to verify each of the following claims:
 - (a) Meta and Appen publicly announced a joint initiative to ensure all Brazilian fact-checkers are paid a fair living wage of at least \$15 USD per hour for combating disaster-related misinformation.
 - (b) The house that the Hamilton College President resides in is called the Kirkland House.
 - (c) The matrix $\begin{pmatrix} 2.5 & 2 \\ 2 & 2.5 \end{pmatrix}$ has eigenvalues $\frac{5}{2}$ and $\frac{1}{2}$.
 - (d) The SMOL language model was trained on a corpus of 100 billion tokens.

PCA and t -SNE by Hand: Two Toy Examples

1. Consider the dataset consisting of four points:

$$(0, -0.2), \quad (2, 0.4), \quad (4, 1.4), \quad (6, 2.4).$$

- (a) Plot the four points in the plane.
- (b) Describe the geometric relationship between x and y .
- (c) Do the points appear to lie approximately along a line?

2. Compute the mean of each coordinate:

$$\bar{x}, \quad \bar{y}.$$

Then subtract the mean from each point to obtain a centered dataset.

Verify that the centered points are:

$$(-3, -1.2), \quad (-1, -0.6), \quad (1, 0.4), \quad (3, 1.4).$$

3. Form the centered data matrix X . Then compute the covariance matrix.

4. Recall that the covariance matrix

$$\begin{pmatrix} \text{Var}(x) & \text{Cov}(x, y) \\ \text{Cov}(x, y) & \text{Var}(y) \end{pmatrix}$$

describes how the dataset spreads in different directions.

- (a) What are the variances of x and y ?
 - (b) Which variable varies more?
 - (c) Is the covariance positive or negative?
 - (d) What does this tell you about how x and y change together?
5. Based on the covariance matrix, predict the direction of the first principal component.
 - (a) Is the dominant direction closer to horizontal, vertical, or diagonal?
 - (b) Explain your reasoning without computing eigenvectors.
 6. PCA identifies directions in which the dataset varies the most.
 - (a) Plot the first principal component direction on the same graph as the data points.
 - (b) Project the data points onto the first principal component direction and plot the projected points on a number line.
 - (c) How much of the variance does the first principal component account for?
 7. Compare this dataset to the example in the notes in which the covariance matrix has equal diagonal entries.
 - (a) Which dataset has a stronger preferred direction of variation?
 - (b) Why does PCA reduce this dataset effectively to one dimension?
 8. Consider the following six points in the plane:

$$A = (0, 0), \quad B = (1, 0), \quad C = (0, 1),$$

$$D = (5, 5), \quad E = (6, 5), \quad F = (5, 6).$$

Plot these six points.

- (a) Which points appear close together?
- (b) Which points appear far apart?
- (c) How many clusters do you see?

9. Compute the squared distances from point A to each of the other points.

That is, compute:

$$d(A, B)^2, \quad d(A, C)^2, \quad d(A, D)^2, \quad d(A, E)^2, \quad d(A, F)^2.$$

Which distances are small? Which are large?

10. In t -SNE, similarities are converted into probabilities using

$$P(j | A) = \frac{e^{-d(A, j)^2}}{\sum_{k \neq A} e^{-d(A, k)^2}}.$$

(a) Explain why $P(B | A)$ and $P(C | A)$ should be large.

(b) Explain why $P(D | A)$, $P(E | A)$, and $P(F | A)$ should be very small.

11. Repeat the same reasoning for point D .

(a) Which points are most likely neighbors of D ?

(b) Which points are unlikely to be neighbors of D ?

12. A t -SNE embedding tries to preserve neighbor relationships rather than exact distances.

(a) Sketch a possible two-dimensional arrangement of the six points that preserves these neighbor relationships.

(b) Should the distance between clusters be interpreted precisely?

(c) Should distances within clusters be interpreted precisely?

13. Compare how PCA and t -SNE would treat this dataset.

(a) Would PCA primarily detect a direction of variation or a clustering structure?

(b) Why is this dataset better suited to t -SNE than PCA?